

연구커뮤니케이션 - 인터뷰, 면접에 대한 발표; 그리고 참고문헌 -

MedAI-Research-5

240928

최기선



Interviews

PhD positions, Internships and Postdoctoral Fellows





What are the interviews for?

- Interviewers want to know if your skills match their need
- It's mutual
 - you need to decide if the job and workplace matches your expectations
 - Ask questions & be curious



www.phdcomics.com

PhD positions

- Check for universities with an active NLP group
- Find supervisors that publish in the NLP venues (e.g., *ACL/EMNLP/COLING/LREC)
- Different continents and countries have their own procedure. Read the information before applying!
 - US
 - Europe
 - Australia
 - Asia
 - Africa

Internships

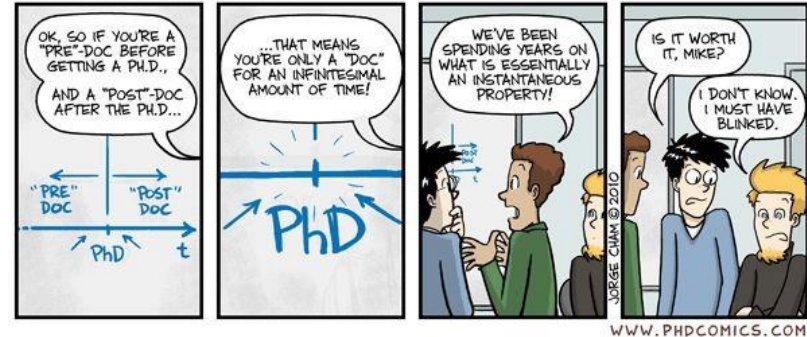
- Short-term jobs
- Gives you a flavour of what to expect after PhD
- Opportunity to work with those researchers you wish to learn from
- Can open more options after graduation
- May lead to one or more publications
- May lead to scholarships



Postdoctoral Fellow (postdoc) or Research Fellow

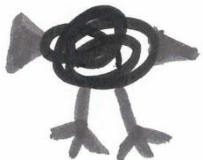
- Early career researchers
- Typically 2-3 years after PhD
 - Same as PhDs, postdoc positions are different in different continents
 - In Australia: Typically 3-years
- One project
- Aim is to get you ready for the next job (industry or academia)

Publish as much as you can!



Applying for jobs or PhD positions

- A current CV/Resume
- A cover letter to state why you wish to be considered for a given position
- Any relevant documents (if asked) such as university qualifications
- A list of references (check with them first, and ask permission)



So, you've been asked to do an interview

Congratulations! It's great to be shortlisted!

- Communicate immediately if there are date and times you are not available.
- Time to prepare!
 - Also known as “do your homework”!
 - Make sure you know the organisation: what they do and why
 - The interviewers background (if known in advance)
 - Some questions to ask about the job!



Interview day

- Be punctual!
- If you cannot make it, let them know ASAP (cancel or rearrange)
- Check in advance that the technology works (online)
- If you're to rely on technology for translation, let them know

Introduce yourself!

It is vital to get this right!

- Explain your PhD in an elevator pitch level
- Be ready to explain your papers, your contributions
- Say what you learnt in the previous position (skills)



1. PhD
2. Honours
3. Masters
4. Internship
5. ...

It's about me!



elevator pitch

- a brief, concise explanation of your work,
- typically lasting 30-60 seconds,
- as if you were summarizing it to someone during a short elevator ride.
- The goal is to convey the essence of your work & research
 - what you're studying,
 - why it's important, and
 - what impact it could have
- — without going into too much technical detail.

elevator pitch – a simple structure

1. Introduction:

- Start with a hook—why is your research important or interesting?

2. Problem Statement:

- What problem are you addressing or solving?

3. Your Approach:

- Briefly explain your research method or approach.

4. Impact:

- Mention the potential outcomes or applications of your research.

Listen to Understand

- Only way to be able to answer a question is by understanding it!
- No need to rush! Ask for clarification! Check your understanding if not sure!



How to answer questions during an interview



- Answer the questions directly and honestly
 - Facts only
 - Examples
 - Have you applied for a PhD admission yet? The answer is yes or no. You can also mention when, outcomes.
 - Do you have a paper published in an *ACL venue? Yes or no. “Thinking about writing” is not an acceptable answer.
- Again, ask for clarifications if unsure
 - It shows you care and you can have a conversation

What if I do not know the answer?

- “I don’t know” is a perfect answer!
- Interviewers would appreciate:
 - Honesty
 - Not having their time wasted with irrelevant fillers





What to avoid during an interview

- Unproductive criticism
 - Criticise the interviewer's research to show you're aware of their work
 - It's not a peer-review process!



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 - What is currently on-trend is not all that matters (respect past work)
 - SVMs can still be useful in some applications
 - LLMs are not the answer to all problems

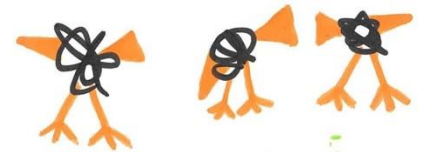


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 - LLMs are not the answer to all problems
- Unwanted slides and materials
 - If the panel is not asking for a presentation, do not insist on that
 - Reliance on prepared material can derail you from listening and answering the questions

Follow ups: More interviews

- Depending on the position, some interviews are followed by
 - Coding tasks
 - Talks to a larger audience
 - Requests for documents or providing evidence
 - References: This is different in different countries (e.g., in Australia usually only if a candidate makes it to a potential hire; in the US done before interviews)



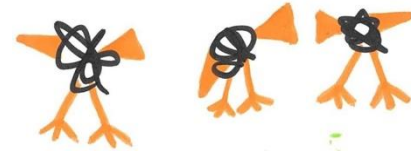
Follow ups: Interview talk



- **Audience:** Ask in advance who will be in the audience
 - Are they all NLP experts? Will they know your area of expertise?
- **Time:** Ask about your allocated time and stick to it! That's respect to their time and schedule!
- **What to Present:** You may want to present a summary of multiple projects or your most prominent one.
 - Slides filled with equations and tables of results may not be as impressive as you may think.
- **They want to know:** are you able to communicate your research well?

Follow ups: Verbal or Written Outcome

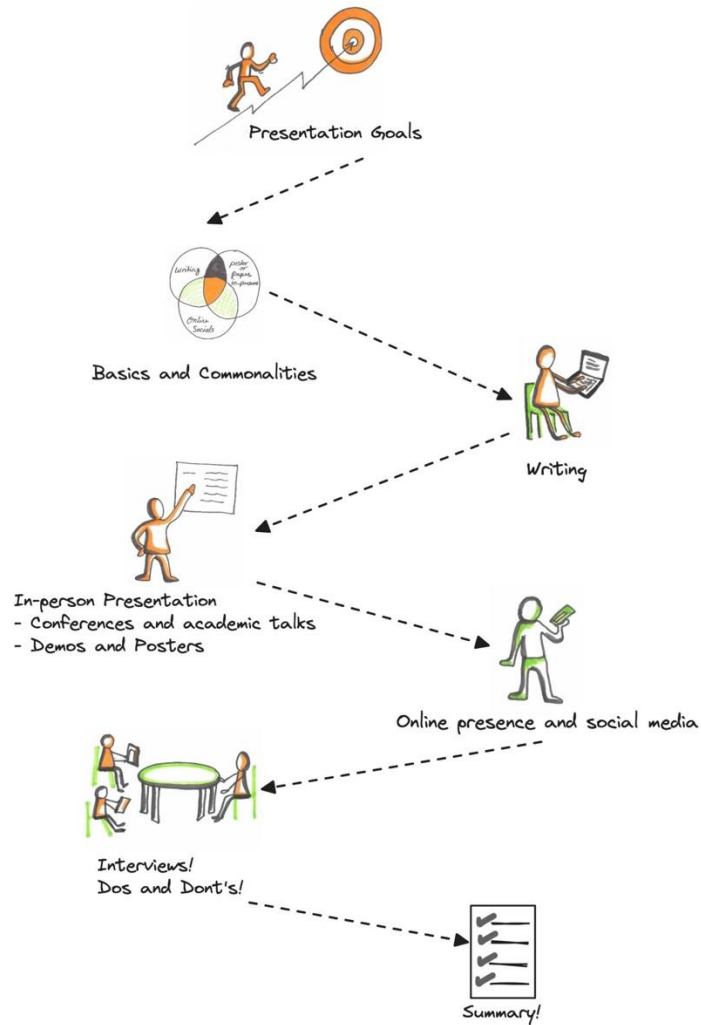
- **Negative:**
 - Keep it professional and polite! You may re-apply or meet them in a conference!
 - Learn from the feedback
- **Positive:**
 - Do acknowledge that you received it
 - Have an answer ready (accept or not) and when can you start



From your experiences of interviews.... Other tips?



Almost there!



Acknowledgements

We are grateful for the materials provided to us by our colleagues:

- Prof. Robert Dale
- Prof Guilin Qi
- Prof. Barbara Plank
- Prof. Wray Buntine
- Dr Simon Peyton Jones
- Dr Carolin Lawrence
- Zhijin Jin and others in the ACL Mentorship program

Summary

1. Know your audience!
2. Have a take home message, story and stick to it!
3. Prepare, prepare, prepare!



Resources

Presentation Checklist

- ☐ Slide numbers
- ☐ Meaningful final slide with maximum of three take home messages
- ☐ All the colours are visible
- ☐ Proofread
- ☐ Each slide has one message

Writing Checklist

- ☐ Re-read the abstract and conclusions to ensure they match the story.
- ☐ Contributions are listed in the introductions.
- ☐ Check all the acronyms are expanded in their first mention in the text.
- ☐ All the tables and figures are referenced in the text.
- ☐ Spell-check and grammar check before submitting.

Social media post Checklist

- ☐ Is this a meaningful content?
- ☐ Is it respectful?
- ☐ Is it professional? Will I be proud of it 5 years from now?
- ☐ Have I proofread properly?
- ☐ Is the media such as photos content mine to post (copyright)?

Interview Checklist

- ☐ Ready to explain my past research (PhD/Master/Honors) in 2-3 sentences
- ☐ Have read about the organisation and know what they do
- ☐ Ready to talk (meaningfully) about my publication(s) in details if they ask
- ☐ Checked all the equipment such as headset works (if online)
- ☐ Questions ready to check if this is the position for me

Resources

Presentations:

How to give a good talk on a Computational Linguistics topic (Barbara Plank)

<https://naacl2018.wordpress.com/2018/05/27/how-to-give-a-good-talk-on-a-computational-linguistics-topic/>

<https://www.cs.jhu.edu/~jason/advice/how-to-give-a-talk.html> (Jason Einser)

<https://en.itu.dk/Research/PhD-Programme/PhD-Courses/PhD-courses-2021/PhD-Course---Communicating-State-of-the-art-NLP-Research-to-a-Broader-Audience>

Covey, S. R. (2004). **The 7 habits of highly effective people (book)**

Resources

Writing:

Writing for Computer Science (Justin Zobel) - Book

Explaining explaining: a quick guide on explanatory writing (Lucas Costa) <https://lucasfcosta.com/2021/09/30/explaining-in-writing.html>

Tips for Writing NLP Papers (Vered Shwartz) <https://medium.com/@vered1986/tips-for-writing-nlp-papers-9c729a2f9e1f>

PhD applications:

PhDs in Europe: <https://soundcloud.com/nlp-highlights/134-phd-application-series-phds-in-europe-versus-the-us>

PhDs in the US:

<https://soundcloud.com/nlp-highlights/133-phd-application-series-preparing-application-materials-with-nathan-schneider-and-roma-patel>

Resources

ACL Mentorship:

- ACL Mentorship YouTube videos: <https://www.youtube.com/watch?v=jFWF0PX4Pcs&list=PLqX2Wb03yR2Sg6eM4xHujbHlyqnvalufy>
- ACL mentorship home page: <https://mentorship.aclweb.org/>

Resources

<http://users.umiacs.umd.edu/~jbg/static/style.html>

<https://www.microsoft.com/en-us/research/academic-program/write-great-research-paper/>

<https://www.youtube.com/watch?v=TCytsY8pdsc>

<https://www.wiley.com/en-us/network/publishing/research-publishing/writing-and-conducting-research/how-to-write-a-scientific-abstract>

<https://www.youtube.com/watch?v=pbrxGnL3tvQ>

<https://aaai.org/aaai-conference/reproducibility-checklist/>

<https://www.youtube.com/watch?v=0h8woOdNSN4>

Resources

[Research is communication \(frankxfz.me\)](https://frankxfz.me/research-is-communication) provides some good ideas for writing a good scientific paper.

[how-to-be-a-successful-phd-student.pdf \(frankxfz.me\)](https://frankxfz.me/how-to-be-a-successful-phd-student.pdf) has good suggestions about when to write.

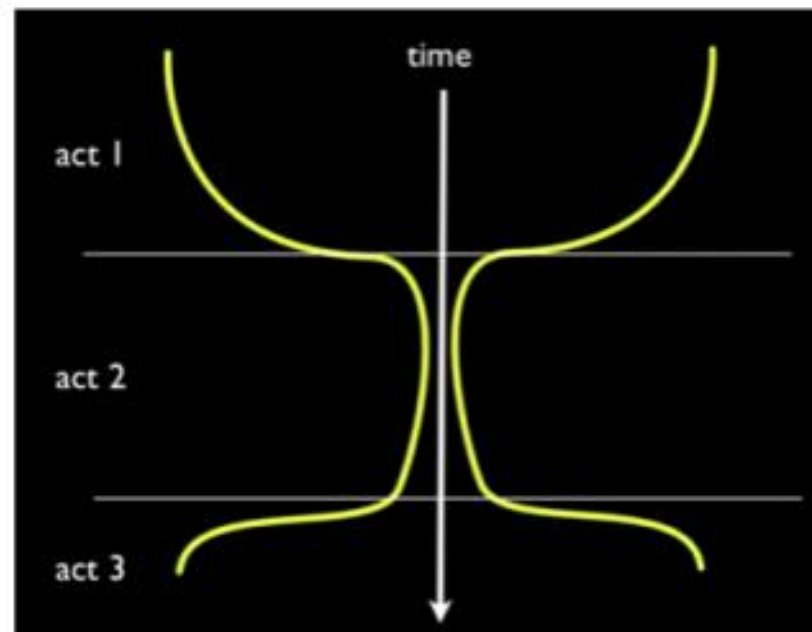
For many NLP projects, it would be great to include one or two sentences summary in the introduction to highlight the key contributions of this work. The reviewers can easily reuse such sentences in peer reviews.

It would be great to understand the peer review process and mitigate some common pitfalls via good writing. [What Can We Do to Improve Peer Review in NLP? - ACL Anthology](#) and the recent white paper [2405.06563 \(arxiv.org\)](#) written by us.

Behavioral consequences of second-person pronouns in written communications between authors and reviewers of scientific papers <https://www.nature.com/articles/s41467-023-44515-1>

The structure of Narrative

- Act 1: beginning (act one) sets the scene and context,
- Act 2: main content,
- Act 3: the resolution and take-away message



<https://naacl2018.wordpress.com/2018/05/27/how-to-give-a-good-talk-on-a-computational-linguistics-topic/>

The structure of Narrative: shape of the glass is determined by the audience

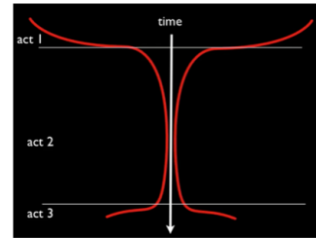
a) **Martini glass:** technical talk - audience is typically from the same field and very familiar with the main concepts. So the bowl can be wide and short.

a) with a narrow stem and more technical details.

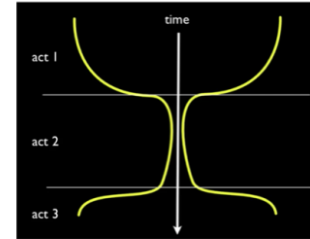
b) **Wine glass:**

c) **Whisky glass:** for public talk - spend more time on setting the scene and providing context.

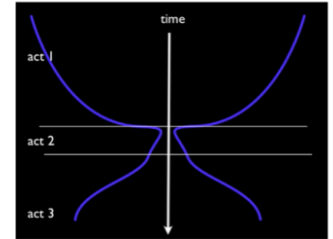
Narrative structure:



(a) technical symposium



(b)



adapted from Tim Miller (2012)

(c) public talk

Audience